

Exercise Sheet 5

Exercise 1 (Memory Management)

1. Mark memory management methods that cause internal fragmentation to occur.
☐ Static partitioning ☐ Dynamic partitioning ☐ Buddy memory allocation
2. Mark memory management methods that cause external fragmentation to occur.
☐ Static partitioning ☐ Dynamic partitioning ☐ Buddy memory allocation
3. Explain how external fragmentation can be fixed.
4. Mark the memory management method that searches in the entire address space for the block, which fits best to satisfy the request.
☐ First Fit ☐ Next Fit ☐ Best fit ☐ Random
5. Mark the memory management concept that searches for the first free block that satisfies the request, starting from the beginning of the address space.
☐ First Fit ☐ Next Fit ☐ Best fit ☐ Random
6. Mark the memory management concept that fragments quickly the large area of free space at the end of the address space.
☐ First Fit ☐ Next Fit ☐ Best fit ☐ Random
7. Mark the memory management concept that selects randomly a free and appropriate block.
☐ First Fit ☐ Next Fit ☐ Best fit ☐ Random
8. Mark the memory management concept that searches for a free block, starting from the latest allocation.
☐ First Fit ☐ Next Fit ☐ Best fit ☐ Random
9. Mark the memory management concept that produces many mini-fragments and is slow.
☐ First Fit ☐ Next Fit ☐ Best fit ☐ Random
10. Static partitioning can only be implemented using partitions of equal size.
☐ True ☐ False

11. The following memory area belongs to a memory with dynamic partitioning. For each of the three algorithms, First Fit, Next Fit, and Best Fit, specify the number of the free partition that the corresponding algorithm uses to insert a process that requires 21 MB of memory.

a) First Fit: _____ b) Next Fit: _____ c) Best Fit: _____

	10 MB	0
	22 MB	1
	30 MB	2
last partition assigned →	2 MB	3
	7 MB	4
	17 MB	5
	12 MB	6
	45 MB	7
	21 MB	8
	39 MB	9

free
occupied

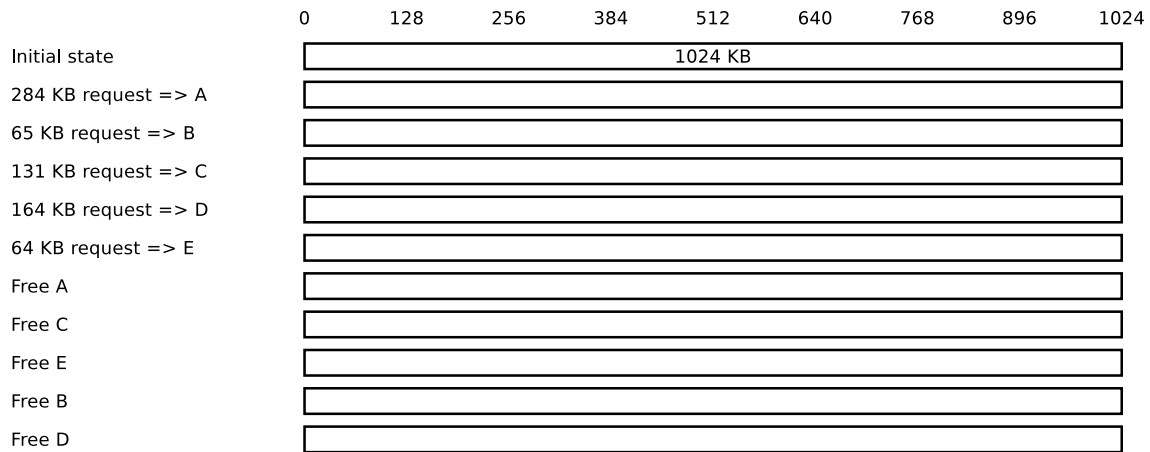
Exercise 2 (Buddy Memory Allocation)

The Buddy method for allocating memory to processes shall be used for a memory with a capacity of 1024 kB. Perform the provided operations and give the occupancy state of the memory after each operation.

	0	128	256	384	512	640	768	896	1024
Initial state	1024 KB								
65 KB request => A									
30 KB request => B									
90 KB request => C									
34 KB request => D									
130 KB request => E									
Free C									
Free B									
275 KB request => F									
145 KB request => G									
Free D									
Free A									
Free G									
Free E									

Exercise 3 (Buddy Memory Allocation)

Apply the Buddy Allocation algorithm to the memory depicted in the diagram.



Exercise 4 (Real Mode and Protected Mode)

1. Describe the functioning of the real mode.
2. Explain why it is impossible to use real mode for multitasking operation mode.
3. Describe the functioning of the protected mode.
4. Describe what virtual memory is.
5. Explain, why virtual memory helps to better utilize the main memory.
6. Describe what mapping is.
7. Describe what swapping is.
8. Name the component of the CPU that is used to implement virtual memory.
9. Describe the function of the component from subtask 8.
10. Describe the virtual memory concept called paging.
11. Describe where internal fragmentation occurs in paging.
12. Give the maximum number of memory addresses a 16-bit computer system can address.
13. Give the maximum number of memory addresses a 32-bit computer system can address.

14. Give the maximum number of memory addresses a 64-bit computer system can address.
15. Explain why multi-level paging is used in 32-bit and 64-bit systems, rather than single-level paging.
16. Calculate the 16-bit physical memory address using single-level paging address conversion. Fill in the individual bits in the 16-bit physical address.

Virtual (logical) 16-bit address

0	0	0	1	0	1	1	1	0	1	1	1	0	1	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

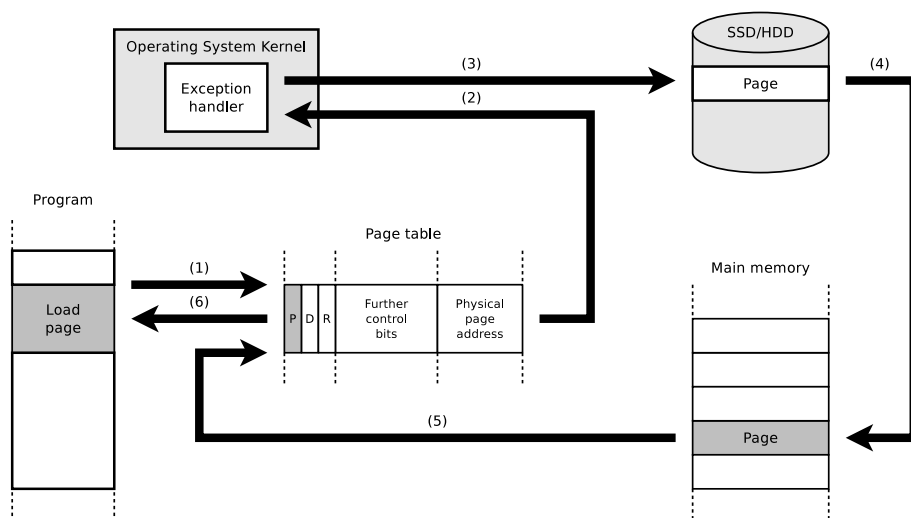
Page table

...															
0 0 0 1 1 0	P	D	R	Further control bits	1	0	0	1	0	1					
0 0 0 1 0 1	P	D	R	Further control bits	1	1	1	0	1	0					
...															
0 0 0 0 1 0	P	D	R	Further control bits	0	0	1	0	1	1					
0 0 0 0 0 1	P	D	R	Further control bits	0	1	1	0	1	1					
0 0 0 0 0 0	P	D	R	Further control bits	0	1	1	1	0	1					

Physical 16-bit address

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

17. Explain the purpose of the Page-Table Base Register (PTBR).
18. Explain the purpose of the Page-Table Length Register (PTLR).
19. Explain the event that causes a page fault exception.
20. The diagram shows a page fault exception. Describe the process stages.



(1)

(2)

(3)

(4)

(5)

(6)

21. Explain what an access violation exception or general protection fault exception causes to occur.
22. Describe the consequence (effect) of an access violation exception or general protection fault exception.
23. Describe the content of the kernelspace.
24. Describe the content of the userspace.

Exercise 5 (Memory Management)

Please mark for each one of the following statements, whether the statement is true or false.

1. Real mode is suited for multitasking systems.

☐ True ☐ False

2. In protected mode, each process is executed in its own copy of the physical address space, which is protected from other processes.

☐ True ☐ False

3. When static partitioning is used, internal fragmentation occurs.

☐ True ☐ False

4. When dynamic partitioning is used, external fragmentation cannot occur.

☐ True ☐ False

5. With paging, all pages have the same length.

☐ True ☐ False

6. One advantage of long pages is little internal fragmentation.

☐ True ☐ False

7. A drawback of short pages is that the page table gets bigger.

☐ True ☐ False

8. When paging is used, the MMU translates the logical memory addresses into physical memory addresses.

☐ True ☐ False

9. Modern operating systems (for x86) operate in protected mode and use only paging.

☐ True ☐ False

Exercise 6 (Page Replacement Strategies)

1. Why is it impossible to implement the optimal replacement strategy OPT?
2. Perform the access sequence with the replacement strategies Optimal, LRU, LFU and FIFO once with a cache with a capacity of 4 pages and once with 5 pages. Also calculate the hit rate and the miss rate for all scenarios.

Optimal replacement strategy (OPT):

Requests: **1 3 5 4 2 4 3 2 1 0 5 3 5 0 4 3 5 4 3 2 1 3 4 5**

Page 1:																							
Page 2:																							
Page 3:																							
Page 4:																							

Hit rate:

Miss rate:

Requests: **1 3 5 4 2 4 3 2 1 0 5 3 5 0 4 3 5 4 3 2 1 3 4 5**

Page 1:																							
Page 2:																							
Page 3:																							
Page 4:																							
Page 5:																							

Hit rate:

Miss rate:

Replacement strategy Least Recently Used (LRU):

Requests: 1 3 5 4 2 4 3 2 1 0 5 3 5 0 4 3 5 4 3 2 1 3 4 5

Page 1:

Page 2:

Page 3:

Page 4:

Queue:

Hitrate:

Missrate:

Requests: 1 3 5 4 2 4 3 2 1 0 5 3 5 0 4 3 5 4 3 2 1 3 4 5

Page 1:

Page 2:

Page 3:

Page 4:

Page 5:

Queue:

Hitrate:

Missrate:

Replacement strategy Least Frequently Used (LFU):

Requests: **1 3 5 4 2 4 3 2 1 0 5 3 5 0 4 3 5 4 3 2 1 3 4 5**

Page 1:																							
Page 2:																							
Page 3:																							
Page 4:																							

Hit rate:

Miss rate:

Requests: **1 3 5 4 2 4 3 2 1 0 5 3 5 0 4 3 5 4 3 2 1 3 4 5**

Page 1:																							
Page 2:																							
Page 3:																							
Page 4:																							
Page 5:																							

Hit rate:

Miss rate:

Replacement strategy FIFO:

Requests: **1 3 5 4 2 4 3 2 1 0 5 3 5 0 4 3 5 4 3 2 1 3 4 5**

Page 1:																							
Page 2:																							
Page 3:																							
Page 4:																							

Hit rate:

Miss rate:

Requests: **1 3 5 4 2 4 3 2 1 0 5 3 5 0 4 3 5 4 3 2 1 3 4 5**

Page 1:																							
Page 2:																							
Page 3:																							
Page 4:																							
Page 5:																							

Hit rate:

Miss rate:

3. Describe the key message of Laszlo Belady's anomaly.

4. Show Belady's anomaly by performing the access sequence with the replacement strategy FIFO once with a cache with a capacity of 3 pages and once with 4 pages. Also calculate the hit rate and the miss rate for both scenarios.

Requests: **3 2 1 0 3 2 4 3 2 1 0 4**

Page 1:											
Page 2:											
Page 3:											

Hit rate:

Miss rate:

Requests: **3 2 1 0 3 2 4 3 2 1 0 4**

Page 1:											
Page 2:											
Page 3:											
Page 4:											

Hit rate:

Miss rate:

Exercise 7 (Time-based Command Execution, Sorting, Environment Variables)

1. Create in your home directory a directory `NotImportant` and write a cron job, which erases the content of the directory `NotImportant` every Tuesday at 1:25 clock am.

The output of the command should be appended to a file `EraseLog.txt` in your home directory.

2. Write a cron job, which appends a line at a file `Datum.txt` with the following format (but with the current values) every 3 minutes between 14:00 to 15:00 clock on every Tuesday in the month of November:

```
Heute ist der 30.10.2008
Die Uhrzeit ist 09:24:42 Uhr
*****
```

3. Write an at-job, which outputs at 17:23 today a list of the running processes.

*You may have to install the command line tool **at** first.*
With Debian/Ubuntu this works with:
\$ sudo apt update && sudo apt install at
With CentOS/Fedora/RedHat this works with:
\$ sudo yum install at

4. Write an at-job, which outputs at December 24th at 8:15 am the text „It’s christmas!“

5. Create in your home directory a file `Kanzler.txt` with the following content:

Willy	Brandt	1969
Angela	Merkel	2005
Gerhard	Schröder	1998
KurtGeorg	Kiesinger	1966
Helmut	Kohl	1982
Konrad	Adenauer	1949
Helmut	Schmidt	1974
Ludwig	Erhard	1963

6. Print out the file `Kanzler.txt` sorted by the first names.
7. Print out the file `Kanzler.txt` sorted by the third letter of the last names.
8. Print out the file `Kanzler.txt` sorted by the year of the inauguration.
9. Print out the file `Kanzler.txt` backward reverse sorted by the year of the inauguration and redirect the output into a file `Kanzlerdaten.txt`.
10. Create with the command `export` an environment variable `VAR1` and assign it the value `Testvariable`.
11. Print out the value of `VAR1` in the shell.
12. Erase the environment variable `VAR1`.